

Answer Key — Habitat and Its Components

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (b) **habitat** — the place an organism lives is its habitat.

Q2 (c) **sunlight** — sunlight is non-living — abiotic.

Q3 (b) **biotic** — living things are biotic components.

Q4 (b) **abiotic** — oxygen/water are abiotic (non-living) needs.

Q5 (b) **False** — soil and water are non-living — abiotic.

Q6 **biotic** — living parts are biotic.

Q7 **abiotic** — non-living parts are abiotic.

Short answer

Q8 The place where an organism lives, e.g. a pond, a forest (or even a tree's bark).

Q9 Biotic = living things (plants, animals, microbes); abiotic = non-living things (water, soil, air, sunlight).

Q10 Biotic needs: food from small plants and animals; abiotic needs: oxygen, water, space and shelter.

Long / application

Q11 Biotic: algae/small animals (food), other organisms. Abiotic: water, oxygen, sunlight, soil. The fish depends on biotic components for food and on abiotic components like water and dissolved oxygen for breathing and living space — both are needed for survival.

Q12 They share the same habitat but use it at different times and in different ways — the rodent is active in the warmer day, the snake at night — so they face different conditions and don't directly compete at the same time, allowing them to coexist in harmony.
